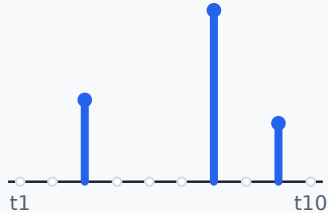


(a) Sparse demand

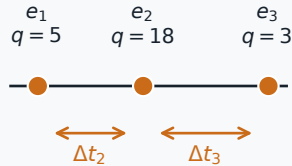
Fixed-grid series contains long zero-demand intervals.



Zero-heavy regular observations

(b) Marked events

Retain positive-demand events and their inter-event times.



History $\mathcal{H}_i = \{(t_j, q_j)\}_{j \leq i}$

(c) Quantity factorization

A categorical mark alone identifies a range, not the exact quantity.

coarse magnitude mark
 $m_i = \min(\lfloor \log_b q_i \rfloor, M)$

continuous residual
 $r_i = \log_b q_i - m_i$

$$q_i = b^{m_i + r_i}$$

Example: $q = 18, b = 2: m = 4, r \approx 0.17$

(d) Next-event prediction

Predict when demand returns and how much demand arrives.

magnitude mark
 $p(m_{i+1} | \mathcal{H}_i)$

event time
 $f(\Delta t_{i+1} | \mathcal{H}_i)$

continuous quantity
 $\hat{r}_{i+1} \rightarrow \hat{q}_{i+1}$

$$\mathcal{L} = \mathcal{L}_{\text{mark}} + \mathcal{L}_{\text{time}} + \mathcal{L}_{\text{res}} + \lambda_q \mathcal{L}_{\text{qty}}$$